

디지털 전자회로(공)(반) (SM307-01)

학기	2026년 2학기
담당교수	김가인
이수구분	심화필수
학점	3.0
강의시간	월14:30-16:00(E3 - 626), 수14:30-16:00(E3 - 626)
강의실	E3 409 (교수실)
소속	반도체공학과

교과개요

This course covers fundamentals of digital integrated circuit (IC) design, transistor-level design of logic cells, transistor-level design of arithmetic elements, sequential circuits, power/area/speed trade-off, and basics of semi-custom digital IC design.

교과목표

The objective of this course is to equip students with a comprehensive understanding of the foundational principles and techniques underlying digital IC design. By the end of this course, students will be proficient in transistor-level design methodologies for both logic and arithmetic cells, understand the behaviour of sequential circuits in transistor-level, and be able to navigate the complexities of power, area, and speed trade-offs in IC design.

수업방법

Delineation of basic principles and examples

학습윤리

No tolerance for cheating or any inappropriate behavior.

교과정책(방침)

Grading will be based on the mid-term, final exam